FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: PAST QUESTIONS

B.Sc. MATHEMATICS HONOURS

MAT5CJ303: COMPLEX ANALYSIS I

Module I: Complex Numbers and the Complex Plane (Zill §1.1–1.5)

	Source: all past papers on file (March 2020 to April 2026)	**Marks as printed**
	Section A: 1 to 3 marks	
	Complex numbers and the complex plane	
1	Fill in the blanks: The locus of the points z satisfying $\|z+2i\|^2 = 2\|i+1\|$ is a/an ———. *March 2020*	**1**
2	Solve for z: $3z - 1 = 2\bar{z}$. *March 2020*	**1**
	Section B: 4 to 7 marks	
	Complex numbers and the complex plane	
3	Which one is bigger: $\big\|\|z_1\| - \|z_2\|\big\|$ or $\|z_1 - z_2\|$? Prove your claim. *March 2020*	**4**
4	Determine the locus of points z in the complex plane satisfying the equation $\|z-3\|/\|z-2\| = 2$. *March 2020*	**7**
	Section C: 10 marks and above	
	Complex numbers and the complex plane	
5	Prove or disprove: $\|z_1+z_2\|^2 + \|z_1-z_2\|^2 = 2\left(\|z_1\|^2 + \|z_2\|^2\right)$ for all complex numbers z_1 and z_2. *March 2020 · part (b) of the question*	**13**

A question marked "part (b)" was set with a second part outside this module; only the part belonging to this module is shown, with the marks printed for the whole question.

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Module II: Complex Functions, Mappings and the Power Functions z^n, $z^{1/n}$ (Zill §2.1, 2.2, 2.4)

Source: all past papers on file (March 2020 to April 2026)	**Marks as printed**
No past questions	
None of the eight past papers on file (March 2020, March 2022 ×2, March 2023 ×2, March 2024, March 2025, April 2026) has a question on this module: complex functions as mappings, the power functions z^n and $z^{1/n}$, and the principal n-th root. The two mapping questions in the papers, the logarithmic mapping (March 2024) and the exponential mapping (March 2025), belong to Module IV and are in that paper. No model questions have been added.	

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MAT5CJ303: COMPLEX ANALYSIS I

Module III: Limits, Continuity, Differentiability, Cauchy–Riemann Equations and Harmonic Functions (Zill §3.1–3.4)

Source: all past papers on file (March 2020 to April 2026)		Marks as printed
Section A: 1 to 3 marks		
Limits and continuity		
1	Evaluate $\lim_{z\to 0} \dfrac{1-z^3}{1+z+z^2}$. *April 2026*	**2**
Differentiability, analytic and entire functions		
2	Define an analytic function and give an example of a function which is not analytic at the origin. *March 2020*	**1**
3	Verify whether $f(z)=\bar{z}/z$ is analytic or not at $z=0$? *March 2020*	**1**
4	A complex function $f(z)$ is analytic at a point $z=z_0$ if ———. *March 2022*	**1**
5	Give an example of a complex function which is differentiable at a point but not analytic at that point. *March 2022*	**1**
6	A point $z=z_0$ is a singular point of a complex function $w=f(z)$ if ———. *March 2022*	**1**
7	Define holomorphic function in a domain D. And give an example for an entire function. *March 2022*	**3**
8	Prove or disprove: if f is differentiable at a point z_0, then f is continuous at that point. *March 2022*	**3**
9	Define singular point and also find the singular points of the function $f(z)=\dfrac{\sin z}{z(1-z)}$. *March 2023*	**2**
10	Define entire function. Give an example. *March 2023*	**2**
11	Show that $f(z)=\bar{z}$ is nowhere differentiable. *March 2024*	**2**
12	Define an analytic function. Give an example. *March 2025*	**2**
13	Define L'Hôpital Rule and using it compute $\lim_{z\to i} \dfrac{z^7+i}{z^{14}+1}$. *March 2025*	**2**
14	Does every continuous functions are differentiable ? Justify your answer. *April 2026*	**2**
15	Does the function $f(z)=\bar{z}$ is analytic ? Justify your answer. *April 2026*	**2**
Cauchy–Riemann equations		
16	An analytic function with constant argument is ———. *March 2022*	**1**
17	Write the polar form of Cauchy–Riemann equations. *March 2022*	**1**
18	Check the differentiability of the function $f(z)=2x^2+y+i\left(y^2-x\right)$. *March 2023*	**2**
19	State a necessary condition for analyticity. *March 2023*	**2**
20	Verify Cauchy–Riemann equations for $f(z)=z^2$. *March 2024*	**2**
21	Write the Cauchy–Riemann equations in polar co-ordinates. *March 2024*	**2**
22	Verify the Cauchy–Riemann equations for $f(z)=\mathrm{Re}(z)$. *March 2025*	**2**
23	Verify the Cauchy–Riemann equations for the function $f(z)=z^2-1$ at the point $z=1-i$. *April 2026*	**2**
24	Write the polar form of the Cauchy–Riemann equation. *April 2026*	**2**
Harmonic functions and harmonic conjugates		
25	Is $u(x,y)=x^2-y^2+xy$ a harmonic function ? Justify your claim. *March 2020*	**1**
26	Define harmonic function with example. *March 2022*	**3**
27	Find the harmonic conjugate of $u=x^2-y^2$. *March 2023*	**2**
28	Prove that $u(x,y)=e^{-x}\sin y$ is harmonic. *March 2023*	**2**
29	Define harmonic function and harmonic conjugate function. *March 2024*	**2**
30	Define harmonic function and show that $u(x,y)=x^3-3xy^2-5y$ is harmonic. *March 2025*	**2**
31	Prove that the two families of curves $u(x,y)=x^2-y^2=c_1$ and $v(x,y)=2xy=c_2$ form an orthogonal system. *March 2025*	**2**

Section B: 4 to 7 marks		
Limits and continuity		
32	Using the definition of continuity show that $\sin z$ is continuous throughout the plane. *March 2020*	**4**
33	Show that the function $f(z) = \bar{z}$ is continuous on the whole complex plane. *April 2026*	**5**
Differentiability, analytic and entire functions		
34	Illustrate entire function by an example. *March 2022*	**4**
35	Show that $f(z) = (\bar{z}+1)^3 - 3\bar{z}$ is nowhere differentiable. *March 2023*	**5**
36	Prove that if f is differentiable at a point z_0 in a domain D then f is continuous at z_0. *March 2023*	**5**
37	Show that the function $f(z) = x + 4iy$ is not differentiable at z. *March 2025*	**5**
38	Define an entire function. Does the function $f(z) = \sin z$ is an entire function ? Justify your answer. *April 2026*	**5**
39	Use L'Hospital's rule find (i) $\lim\limits_{z\to i} \dfrac{1+z^6}{1+z^{10}}$, (ii) $\lim\limits_{z\to 0} \dfrac{1-\cos z}{z^2}$. *April 2026*	**5**
Cauchy–Riemann equations		
40	Verify Cauchy–Riemann equations for the function $f(z) = z^3$. *March 2020*	**4**
41	Show that the function $f(z) = \sqrt{xy}$ is not analytic at the origin, even though Cauchy Riemann equations are satisfied at that point. *March 2020*	**7**
42	Prove the formulas for conversion Cauchy–Riemann equation into the corresponding polar form in detail. *March 2020*	**7**
43	Show that the function $f(z) = \sqrt{xy}$ is not analytic at the origin, even though Cauchy Riemann equations are satisfied at that point. *March 2022*	**7**
44	Find the real constants a, b, c and d so that $f(z) = (3x - y + 5) + i(ax + by - 3)$ is analytic. *March 2023*	**5**
45	Verify the Cauchy Riemann equations for $f(x+iy) = \dfrac{x-iy}{x^2+y^2}$. *March 2024*	**5**
46	If $f(z) = u + iv$ then show that $\lvert f'(z)\rvert^2 = u_x^2 + v_x^2 = u_y^2 + v_y^2$. *March 2024*	**5**
47	Prove that an analytic function $f(z) = u + iv$ is constant if its (a) real part is constant; and (b) its modulus is constant. *March 2025*	**5**
48	Suppose that $f(z)$ and $\overline{f(z)}$ are analytic in a domain D. Show that $f(z)$ is constant in D. *April 2026*	**5**
Harmonic functions and harmonic conjugates		
49	Find the analytic function $f(z)$ for which $u(x,y) = \mathrm{Re}(f(z)) = e^x(x\cos y - y\sin x)$. You should express $f(z)$ finally only in terms of z. *March 2020*	**7**
50	If $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then prove that its component functions are harmonic in D. *March 2022*	**7**
51	Find the analytic function $f(z)$ in terms of z, if $u(x,y) = \mathrm{Re}(f(z)) = e^x(x\cos y - y\sin y)$. *March 2022*	**7**
52	Check whether the function U is harmonic or not; if so find its harmonic conjugate: $U(x,y) = x^3 - 3xy^2 - 5y$. *March 2022*	**6**
53	What is the value of the integer n if $x^n - y^n$ is harmonic ? *March 2023*	**5**
Section C: 10 marks and above		
Cauchy–Riemann equations		
54	State and prove Cauchy–Riemann equation. Also state the sufficient condition for differentiability. *March 2022*	**10**
55	State and prove Cauchy Riemann equations. *March 2025*	**10**
Harmonic functions and harmonic conjugates		
56	Verify that the function $u(x,y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane and find the harmonic conjugate function. *March 2024*	**10**
57	Show that $u = x^2 - y^2 - x + 1$ is harmonic. Find a harmonic conjugate v of u such that $u + iv$ is analytic. *April 2026*	**10**

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MAT5CJ303: COMPLEX ANALYSIS I

Module IV: Exponential, Logarithmic, Trigonometric and Hyperbolic Functions (Zill §4.1, 4.3)

	Source: all past papers on file (March 2020 to April 2026)		**Marks as printed**
	Section A: 1 to 3 marks		
	Complex exponential function		
1	Fill in the blanks: $f(z) = e^z$ is periodic with period $=$ ———.	*March 2022*	**1**
2	The solution of the equation $e^z = -3$ is ———.	*March 2022*	**1**
3	Prove that $\overline{e^z} = e^{\bar{z}}$.	*March 2023*	**2**
4	Find all values of z satisfying the equation $e^{z-1} = -ie^3$.	*March 2023*	**2**
5	Solve $e^w = -2$.	*March 2024*	**2**
6	Find the derivative of $e^{z^2-(1+i)z+3}$.	*March 2025*	**2**
	Complex logarithm		
7	Fill in the blanks: The real part of $\log(2z)$ is ———.	*March 2020*	**1**
8	Find the Principal value of $\operatorname{Ln}(1-i)$.	*April 2026*	**2**
	Complex trigonometric functions		
9	Prove that $\sin^2 z + \cos^2 z = 1$.	*March 2022*	**3**
10	Find the real and imaginary parts of $\sin(\bar{z})$.	*March 2023*	**2**
11	Express $\cos(2-4i)$ in the form $a+ib$.	*March 2024*	**2**
12	Find the value of $\cos i$ and $\sin(2+i)$.	*March 2025*	**2**
	Complex hyperbolic functions		
13	Fill in the blanks: The real part of $\sinh(2z)$ is ———.	*March 2022*	**1**
14	Prove that $\cosh^2 z - \sinh^2 z = 1$.	*March 2023*	**2**
15	Show that $\cosh^2 z - \sinh^2 z = 1$.	*April 2026*	**2**
	Section B: 4 to 7 marks		
	Complex exponential function		
16	Find all solutions of $e^z = 2$.	*March 2020*	**4**
17	Find all the solution of $e^z = 2$.	*March 2022*	**4**
18	Briefly explain the exponential mapping $w = e^z$.†	*March 2025*	**5**
	Complex logarithm		
19	Compute the principal value of the complex logarithm $\operatorname{Ln} z$ for $z = i$ and $z = 1+i$.	*March 2023*	**5**
20	Find the image of the annulus $2 \le \lvert z \rvert \le 4$ under the mapping $w = \operatorname{Ln} z$.†	*March 2024*	**5**
	Complex trigonometric functions		
21	Find the real and imaginary parts of the function $f(z) = \sin(z)$.	*March 2020*	**4**
22	Show that $f(z) = \sin z$ is analytic for all z.	*March 2022*	**4**
23	Find all the solutions of the equation $\sin z = 5$.	*March 2022*	**6**
	Complex hyperbolic functions		
24	Does the equation $\cosh 2z = 2$ has a solution ? If so find the solution.	*April 2026*	**5**
	Section C: 10 marks and above		
	Complex trigonometric functions		
25	Prove or disprove: $\lvert \cos z \rvert \le 1$ for all complex numbers z. Justify your claim.	*March 2022 · part (b) of the question*	**13**
26	Find all the solutions of the equation $\sin z = 5$. Also find the modulus of $\sin z$.	*March 2023*	**10**

† Module IV topic, but the part of the book it comes from (the exponential mapping after Periodicity, and the logarithmic mapping after Example 4 in Zill §4.1) lies just beyond the syllabus limits of items 19 and 20. Kept because the question was asked.

A question marked "part (b)" was set with a second part outside this module; only the part belonging to this module is shown, with the marks printed for the whole question.

COMPLEX ANALYSIS I (MAT5CJ303): PAST QUESTIONS

What is in these four papers

Module	Questions	Section A	Section B	Section C
Module I	5	2	2	1
Module II	0	0	0	0
Module III	57	31	22	4
Module IV	26	15	9	2
Total	**88**			

Papers used (descriptive papers only): March 2020 (MAT 6B 10, 2014–18 admission, 120 marks), March 2022 (MAT 6B 10, 120 marks), March 2022 (MTS 6B 11, 2019 admission, 80 marks), March 2023 (MTS 6B 11, 2019 admission), March 2023 (MTS 6B 11, 2020 admission), March 2024, March 2025 and April 2026 (MTS 6B 11, 2020 admission onwards, 80 marks). The two copies of the March 2023 (2020 admission) paper are one paper. The five MCQ-only papers for SDE candidates are left out because their format differs.

Bands: Section A is 1 to 3 marks, Section B is 4 to 7 marks, Section C is 10 marks and above. Inside each band, questions are sorted by topic (in syllabus order), then by year and month. The grey label at the right of each question is the examination month and year. Wording and marks are as printed.

Questions left out: the past papers were set on the earlier CBCSS Complex Analysis course (Sixth Semester), which also had contour integration, Cauchy's theorems, sequences and series, Taylor and Laurent series, singularities and residues, and complex powers and inverse trigonometric functions. Those topics are not in Modules I to IV of MAT5CJ303 and are not in these papers.